

STUDENT RECORD MANGEMENT SYSTEM

SRMS PROJECT

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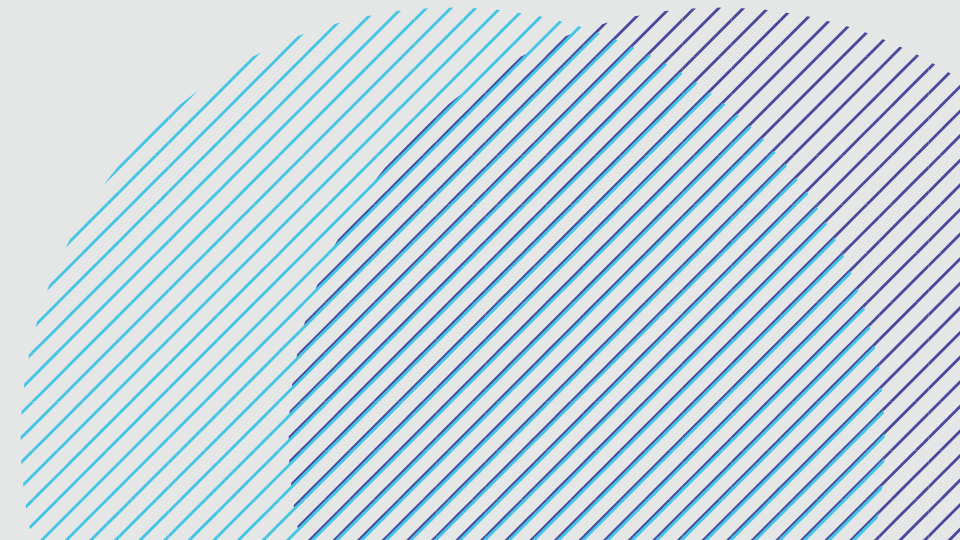
PROJECT OVERVIEW

The system manages university student records (UG & Graduate) using java oop principles main function :

- 1. Add , Update , Delete student**
- 2. Save / load records form CSV**
- 3. Auto-save + Report generation**



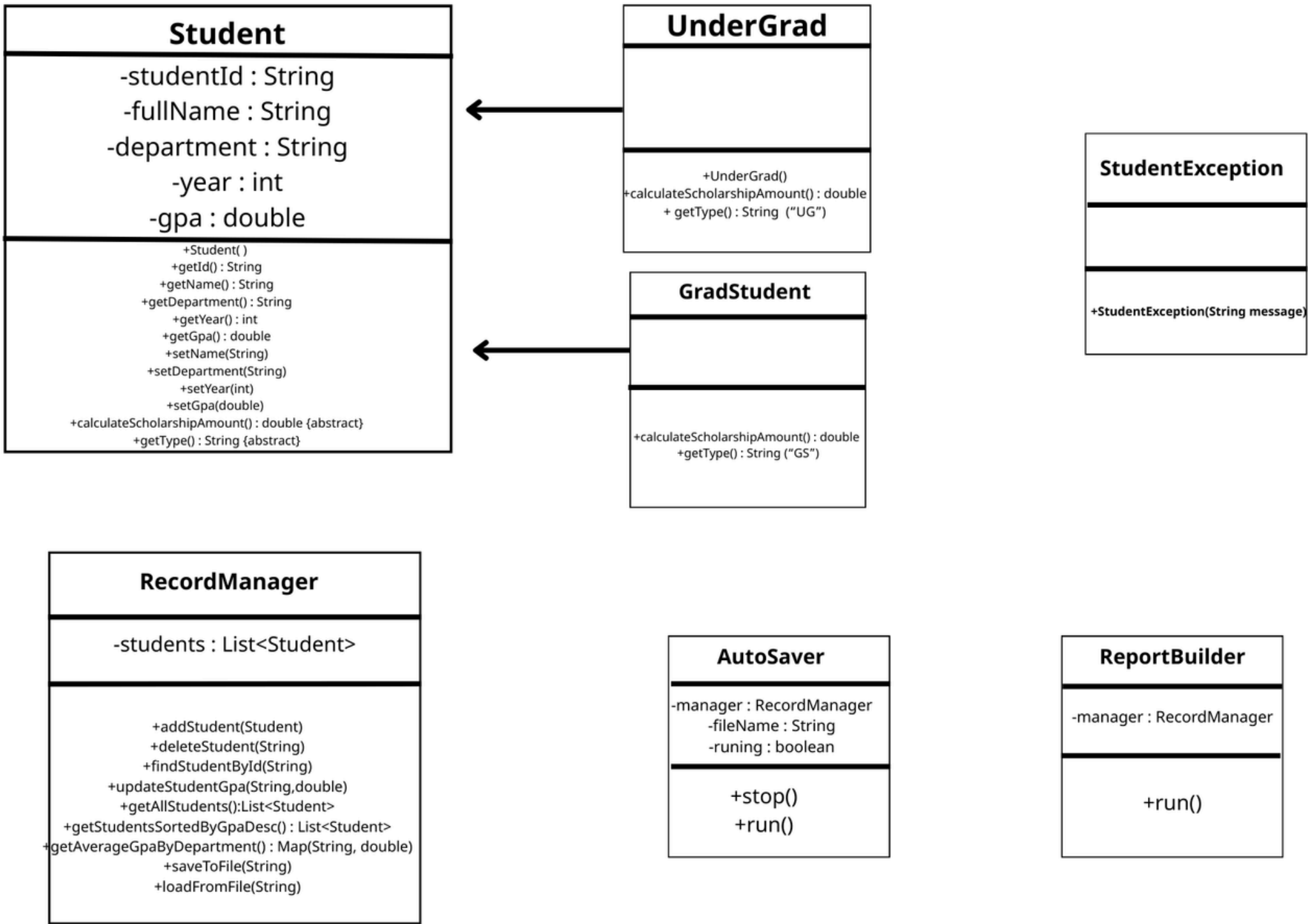
OOP CONCEPTS USED

- **Inheritance : UG & GradStudent extend Student**
 - **Polymorphism : Different scholarship calculation**
 - **Abstraction : Student is abstract**
 - **Encapsulation : Private attributes + getters / setters**
 - **Exception Handling : custom StudentException**
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SYSTEM FEATURES

- *Manage student information*
- *GPA update*
- *Sort by GPA*
- *calculate department GPA averages*
- *Auto-save using multithreading*
- *Generate detailed reports*

UML DIAGRAM



MULTITHREADING

**AutoSaver Thread : Saves data
automatically during runtime**

1

**ReportBulider Thread: Generates
reports whithout stopping
the program**

2

FILE HANDLING

USES CSV FOR :

**Loading data
at
startup**

**Saving manually
and
automatically**

**Keeping records
persistent**



conclusion

The SRMS system demonstrates:

- **Proper use of advanced java OOP**
- **Effective file handling**
- **Working multithreading**
- **Clear design & UML structure**

THANK YOU !